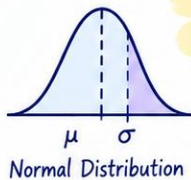


Data
Understand
Inspire
Better Decisions
:)



Concepts
Formulas
Examples
Visuals
All in One Place!

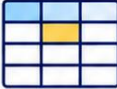


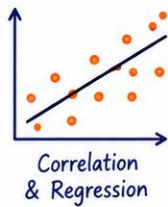
STATISTICS — for — DATA SCIENCE

Σ
Hypothesis
Testing


Confidence
Intervals


Sampling &
CLT


Descriptive
Statistics



FULL NOTES

Clear Concepts • Practical Examples
Real-World Applications

Learn
Apply
Grow 

— DataXOdyssey —

Same Numbers
Bigger Perspectives!


Small
data
Big
decisions!

Statistics Fundamentals

"Data gives you information,
Statistics gives you insights!"

1. What is Statistics?

- Statistics is the science of collecting, organizing, analyzing, interpreting and drawing conclusions from data.
- It helps us make sense of uncertainty, and variability in the real world.

2. Why is it Important in Data Science?

- ✓ Helps in understanding and summarizing data
- ✓ Enables data-driven decision making
- ✓ Useful in building and evaluating ML models
- ✓ Helps in handling uncertainty and making predictions
- ✓ Foundation for experiments, A/B testing and causal analysis

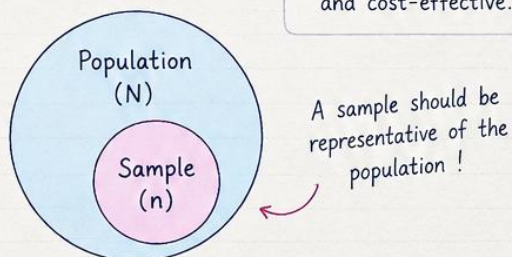
3. Population vs Sample

Population

- The entire group of individuals, items or events of interest.
- Described using parameters (e.g. μ, σ).
- Usually large and often not feasible to study completely.

Sample

- A smaller subset of the population.
- Described using statistics (e.g. $\bar{x}, s$).
- Used to make inferences about the population.
- More practical and cost-effective.



4. Parameter vs Statistic

Parameter

- Numerical value that describes a population.
- Examples: population mean (μ), population standard deviation (σ).
- Fixed (but usually unknown).

Statistic

- Numerical value that describes a sample.
- Examples: sample mean ($\bar{x}$), sample standard deviation (s).
- Varies from sample to sample.
- Used to estimate population parameters.

5. Descriptive vs Inferential Statistics

Descriptive Statistics

- Summarizes and describes the main features of data.
- Uses tables, graphs and summary measures.
- Examples: mean, median, standard deviation, charts.

Inferential Statistics

- Uses sample data to make inferences about a population.
- Involves probability and hypothesis testing.
- Examples: confidence intervals, hypothesis tests, regression analysis.

6. Key Concepts

Data : Raw facts and figures.

Variable : A characteristic that can take different values.

Observation : A single recorded value.

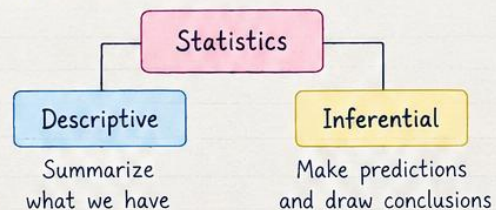
Parameter : Numerical measure of a population.

Statistic : Numerical measure of a sample.

Distribution : Pattern of how data values are spread.

Inference : Drawing conclusions about a population using sample data.

7. Types of Statistics



8. Real Life Example

A company wants to know the average salary of all its employees (population). It surveys 200 employees (sample) and uses the sample mean to estimate the population mean.

"Statistics turns data into understanding" 😊

Understand your data before you analyse it!

Types of Data & Variables

Different types of data need different approaches!

1. What are Variables?

- A **variable** is a characteristic or attribute that can take different values across individuals, items or observations.
- Variables are used to store and represent data.
- Example:** age, gender, income, customer type, temperature, etc.

2. What is Data?

- Data** are the values or observations collected for variables.
- It can be numbers, categories, text, images, etc.
- Data can be **structured** (e.g. tables) or **unstructured** (e.g. text, images).
- Example:** 25, Male, 50000, "Delhi", 3.8, Yes



Data comes in many forms!

3. Types of Data (Broadly)

Qualitative (Categorical)

- Non-numerical data.
- Represents categories or groups.
- Answers "what type" rather than "how much".
- Example:** gender, colour, city, product type.

Quantitative (Numerical)

- Numerical data.
- Represents measurable quantities.
- Answers "how much" or "how many".
- Example:** age, income, height, temperature.

4. Numerical Data Types

Discrete

- Takes distinct, countable values (usually integers).
- Results from counting.
- Example:** number of customers, number of purchases, number of children.

Countable values
(0, 1, 2, 3, ...)

Continuous

- Can take any value within a range (including decimals).
- Results from measurement.
- Example:** height, weight, temperature, time, income.

Any value in a range
(e.g. 0.5, 1.2, 3.8, ...)

5. Categorical Data Types

Nominal

- Categories with no inherent order.
- Example:** gender (Male/Female), city (Delhi, Mumbai, ...), product type.

Ordinal

- Categories with a meaningful order.
- Example:** education level (High School < Graduate < Postgraduate), customer satisfaction (Low < Medium < High).

Nominal vs Ordinal

Both are categorical, but ordinal has a natural order.

Order matters in ordinal!

6. Scales of Measurement

Scale	Description	Example
Nominal	Categories, no order	Gender, City
Ordinal	Categories with order	Education level, Satisfaction level
Interval	Numeric, equal intervals, no true zero	Temperature (°C), Calendar years
Ratio	Numeric, equal intervals, true zero	Height, Weight, Income, Age

7. Examples

Variable	Type	Data Type	Scale
Age	Numerical	Continuous	Ratio
Number of purchases	Numerical	Discrete	Ratio
Gender	Categorical	Nominal	Nominal
Customer rating	Categorical	Ordinal	Ordinal
Income	Numerical	Continuous	Ratio
City	Categorical	Nominal	Nominal
Temperature (°C)	Numerical	Continuous	Interval

8. Key Takeaways

- Identify the **type of data** before choosing analysis methods.
- Categorical and numerical data require different techniques.
- Understanding the **scale** helps in selecting the right statistical tests and visualizations.

"The right analysis starts with understanding your data." 😊

Summarize the data, Visualize the story!

Data Summarization & Visualization

Turning Data into Insights

A picture is worth a thousand rows! 😊

1. Why Summarize & Visualize?

- Helps to understand the main characteristics of the data.
- Simplifies large datasets.
- Reveals patterns, trends, outliers and relationships.
- Helps in making data-driven decisions.
- First step in any data analysis / ML project.

2. Summary Statistics

Measure	What it tells you
Count (n)	Number of observations
Mean	Average value
Median	Middle value (robust to outliers)
Mode	Most frequent value
Min / Max	Smallest / largest value
Range	Max - Min
Variance	Average squared deviation
Standard Deviation	Typical spread of data
Percentiles / Quartiles	Values below which a % of data falls
IQR	Spread of middle 50% (Q3 - Q1)

Use these measures to get a quick understanding of your data!

3. Frequency Table

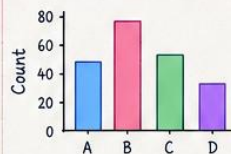
Shows how often each value (or category) occurs.

Category	Frequency	Percentage
A	50	25%
B	80	40%
C	45	22.5%
D	25	12.5%

Useful for categorical data (such as gender, city, product type).

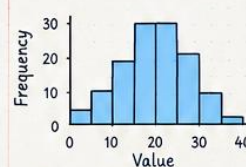
4. Common Visualizations

Bar Chart (categorical data)



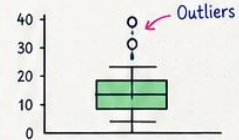
- Compare counts across categories.

Histogram (numerical data)



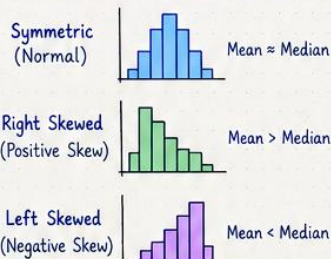
- Shows distribution of a numerical variable.
- Helps identify shape, spread and outliers.

Box Plot (summary + outliers)



- Shows median, quartiles, spread and outliers.
- Useful for comparing distributions across groups.

5. Shape of Data Distribution



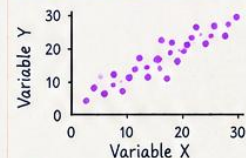
The shape of the distribution helps in choosing the right statistical methods.

Line Chart (trends over time)



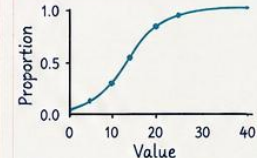
- Shows how a variable changes over time.
- Useful for time series data.

Scatter Plot (relationship between variables)



- Shows relationship between two numerical variables.
- Helps identify correlation, clusters and outliers.

ECDF / Distribution Plot



- Shows the cumulative distribution of data.
- Useful to understand the overall shape and compare distributions.

6. Choosing the Right Plot

Goal	Recommended Plot(s)	Data Type
Compare categories	Bar chart, Count plot	Categorical
See distribution	Histogram, Box plot, ECDF	Numerical
Check relationship	Scatter plot	Numerical vs Numerical
Show trend over time	Line chart	Time series
Compare distributions across groups	Box plot, Violin plot	Numerical (grouped)
See part-to-whole	Pie chart (use sparingly)	Categorical

7. Key Takeaways

- ✓ Use summary statistics to understand the data.
- ✓ Choose the right visualization based on the data type and goal.
- ✓ Look for patterns, trends, outliers and relationships.
- ✓ Good visualization makes insights easy to communicate.

"Summarize the facts, visualize the patterns, make better decisions."

Different datasets, same story... find the center!

Measures of Central Tendency

A single number to represent the center of your data

"Central tendency gives us a quick idea of where the data is centered." 😊

1. What is Central Tendency?

- A central tendency is a single value that represents the "center" or typical value of a dataset.
- Helps to summarize large amounts of data with one number.
- Common measures: Mean, Median, Mode (and sometimes Weighted Mean).

2. When to Use Which?

Measure	Use when...	Best for...
Mean	- Data is numerical - No extreme outliers	Symmetric data (e.g. height, score)
Median	- Data has outliers - Data is skewed	Skewed data (e.g. income, house prices)
Mode	- Data is categorical or discrete - Find the most frequent value	Categorical data (e.g. favourite color) or discrete data (e.g. most common rating)

3. Mean (Arithmetic Mean)

- The sum of all values divided by the number of values.
- Uses every data point.

$$\text{Mean } (\bar{x}) = \frac{\sum_{i=1}^n x_i}{n}$$

Example:

Data: 2, 4, 6, 8, 10

$$\begin{aligned} \text{Mean} &= \frac{2 + 4 + 6 + 8 + 10}{5} \\ &= \frac{30}{5} = 6 \end{aligned}$$

- ✓ Considers all values
- ✗ Sensitive to outliers

4. Median

- The middle value when data is arranged in ascending order.
- Less affected by outliers compared to the mean.

Steps:

- Arrange data in ascending order.
- If n is odd → middle value.
- If n is even → average of two middle values.

Example 1 (Odd):

Data: 3, 7, 8, 12, 15

Median = 8

Example 2 (Even):

Data: 1, 4, 6, 8, 10, 14

$$\text{Median} = \frac{6 + 8}{2} = 7$$

- ✓ Robust to outliers
- ✗ Does not use all values

5. Mode

- The value that appears most frequently in the data.
- Can be used for both numerical and categorical data.

Example 1 (Numerical):

Data: 1, 2, 2, 3, 4, 4, 4, 5

Mode = 4 (appears 3 times)

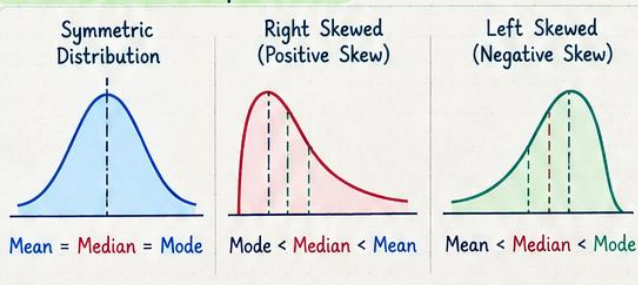
Example 2 (Categorical):

Data: Red, Blue, Red, Green, Red, Blue

Mode = Red (appears 3 times)

- ✓ Simple to understand
Works well for categorical data
- ✗ May have multiple modes or no mode

6. Visual Comparison



7. Key Takeaways

- ✓ Central tendency gives a single number to represent the center of data.
- ✓ Mean uses all values but is sensitive to outliers.
- ✓ Median is robust to outliers.
- ✓ Mode is useful for categorical data.
- ✓ Choice depends on the data distribution and the analysis goal.

"Same data, different center, different insights!"

Same average
can hide
different stories...
Variability shows
the full picture !

Measures of Variability

How spread out is your data ?

"It's not just
about the center,
but also how
much the data
varies." 😊

1. What is Variability ?

- Variability (or dispersion) measures how **spread out** the data is.
- It tells us how much the values **differ from the center** (mean).
- A **low variability** means data points are close to the center.
- A **high variability** means data points are more spread out.

2. Common Measures of Variability

- Range
- Variance
- Standard Deviation
- Mean Absolute Deviation (MAD)
- Coefficient of Variation (CV)

"Different measures,
same goal...
understand the
spread !

3. Example Data

Data: 2, 4, 6, 8, 10
(n = 5)

Mean ($\bar{x}$) = 6

4. Range

- Difference between the maximum and minimum value.
- Simple but only uses two values (ignores the rest).

Range = Max - Min

Example:

Max = 10, Min = 2

Range = 10 - 2 = 8

5. Variance

- Average of the squared differences from the mean.
- Gives more weight to larger differences.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Example:

$$\sum (x_i - 6)^2 = 16 + 4 + 0 + 4 + 16 = 40$$

$$s^2 = \frac{40}{5 - 1} = 10$$

6. Standard Deviation

- Square root of the variance.
- Tells us the typical distance of data points from the mean.
- In the same units as the data.

$$s = \sqrt{s^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Example:

$$s = \sqrt{10} \approx 3.16$$

7. Mean Absolute Deviation (MAD)

- Average of the absolute differences from the mean.
- Less sensitive to outliers than standard deviation.

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

Example:

$$|2 - 6| + |4 - 6| + |6 - 6| + |8 - 6| + |10 - 6| = 16$$

$$MAD = \frac{16}{5} = 3.2$$

8. Coefficient of Variation (CV)

- Relative measure of variability.
- Useful for comparing the spread between different datasets (with different units or means).

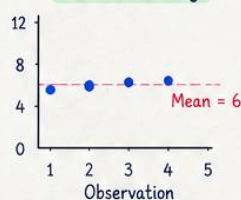
$$CV = \frac{s}{\bar{x}} \times 100\%$$

Example:

$$CV = \frac{3.16}{6} \times 100\% \approx 52.7\%$$

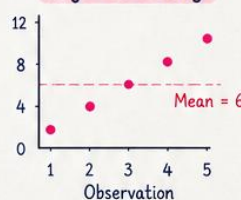
9. Visual Comparison

Low Variability



Data: 4, 5, 6, 7, 8
(Less spread)

High Variability



Data: 2, 4, 6, 8, 10
(More spread)

Both datasets
have the same
mean (6),
but very different
variability !

10. Key Takeaways

- ✓ Variability tells us how spread out the data is.
- ✓ Use range for a quick check, but prefer standard deviation or MAD for better insights.
- ✓ Standard deviation is widely used but sensitive to outliers.
- ✓ MAD is more robust to outliers.
- ✓ Coefficient of Variation helps compare variability across different datasets.
- ✓ Always look at both the center (mean/median) and variability to understand the data fully.

"Same average.
Different story.
Look at the spread !" 😊

"Percentiles help you see where a value stands in the bigger picture!"

Percentiles, Quartiles & Outliers

Understanding the position and unusual values in your data

"Same data, more insights when you look beyond the average!" 😊

1. What are Percentiles?

- A **percentile** tells you the value below which a certain percentage of data falls.
- Divides the data into 100 equal parts.
- The p^{th} percentile is the value below which $p\%$ of the observations lie.
- Example:** 90th percentile means 90% of the data is less than or equal to that value.

2. What are Quartiles?

- Quartiles are special percentiles that **divide the data into 4 equal parts**.
- $Q_1 = 25^{\text{th}}$ percentile (lower quartile)
- $Q_2 = 50^{\text{th}}$ percentile (median)
- $Q_3 = 75^{\text{th}}$ percentile (upper quartile)
- Help us understand the spread and identify potential outliers.

3. Common Percentiles

Percentile	Meaning
0 th	Minimum value
25 th (Q_1)	25% of data is below this value
50 th (Q_2)	Median (middle value)
75 th (Q_3)	75% of data is below this value
100 th	Maximum value

4. Finding Quartiles (Example)

Consider the sorted data:

Data: 3, 5, 7, 8, 12, 15, 18, 20, 25

- Q_2 (Median) = 12 (middle value)
- Q_1 = median of lower half
(3, 5, 7, 8) = (5 + 7) / 2 = 6
- Q_3 = median of upper half
(15, 18, 20, 25) = (18 + 20) / 2 = 19

So, $Q_1 = 6$, $Q_2 = 12$, $Q_3 = 19$

5. Interquartile Range (IQR)

- The Interquartile Range (IQR) measures the spread of the middle 50% of the data.

$$\text{IQR} = Q_3 - Q_1$$

- In the above example:

$$\text{IQR} = 19 - 6 = 13$$

- A larger IQR means more variability in the middle 50%.
- Less affected by extreme values (outliers).

6. Identifying Outliers

- Outliers are data points that are **unusually small or large** compared to the rest of the data.
- Common method (IQR rule):

$$\text{Lower bound} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper bound} = Q_3 + 1.5 \times \text{IQR}$$

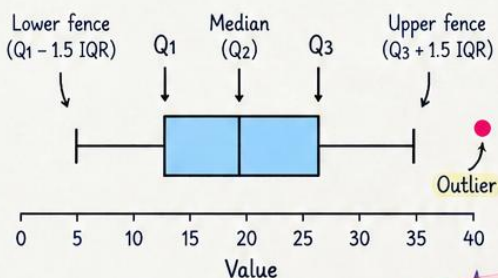
- Using our example:

$$\text{Lower bound} = 6 - 1.5(13) = -13.5$$

$$\text{Upper bound} = 19 + 1.5(13) = 38.5$$

- Any value < -13.5 or > 38.5 is considered an outlier.

7. Visualizing with a Box Plot



- The box shows the middle 50% (IQR).
- The line inside the box is the median (Q_2).
- Whiskers typically extend to $1.5 \times \text{IQR}$.
- Points beyond the whiskers are outliers.

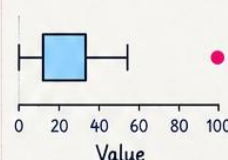
Box plots make outliers easy to spot!"

8. Example with Outliers

Data: 5, 7, 8, 12, 15, 18, 20, 25, 100

- $Q_1 = 7.5$ $Q_3 = 22.5$ $\text{IQR} = 15$
- Lower bound = $7.5 - 1.5(15) = -15$
- Upper bound = $22.5 + 1.5(15) = 45$
- Since $100 > 45$, 100 is an outlier.

Box Plot with Outlier



9. Key Takeaways

- ✓ Percentiles show the relative position of a value in the data.
- ✓ Quartiles divide the data into four equal parts.
- ✓ IQR measures the spread of the middle 50% and is robust to outliers.
- ✓ Use the IQR rule to identify potential outliers.
- ✓ Box plots are a great way to visualize percentiles, quartiles and outliers.
- ✓ Outliers can indicate errors, rare events or important insights — always investigate them!

"Same numbers. Different story!" 😊

"Probability helps us make better decisions in an uncertain world!"

Probability Fundamentals

Quantifying uncertainty in a systematic way

"It's not about knowing what will happen, but how likely it is!"

1. What is Probability?

- Probability is a numerical measure of how likely an event is to occur.
- It takes values between 0 and 1.
- 0 means impossible event.
- 1 means certain event.
- The higher the probability, the more likely the event.

$$0 \leq P(\text{Event}) \leq 1$$

2. Key Terminology

- Experiment:** A process that produces an outcome.
- Outcome:** A possible result of an experiment.
- Sample Space (S):** Set of all possible outcomes.
- Event (A):** A subset of outcomes from the sample space.

3. Example: Rolling a Die

- Experiment:** Roll a fair six-sided die.
- Sample Space:**
 $S = \{1, 2, 3, 4, 5, 6\}$
- Example Events:**
 $A = \{\text{get an even number}\} = \{2, 4, 6\}$
 $B = \{\text{get a number} > 4\} = \{5, 6\}$
 $C = \{\text{get a 7}\} = \emptyset$ (impossible event)



4. Classical Definition of Probability

- Used when all outcomes are equally likely.

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

Example:

A = getting an even number when rolling a die.

$$P(A) = \frac{3}{6} = \frac{1}{2} = 0.5$$

5. Sample Space Examples

- Coin toss:
 $S = \{H, T\}$
- Two coin tosses:
 $S = \{HH, HT, TH, TT\}$
- Drawing a card (52 cards):
 $S = \{\text{all 52 unique cards}\}$
- Drawing a red card:

$$P(\text{Red}) = \frac{26}{52} = \frac{1}{2} = 0.5$$



6. Types of Events

- Simple Event:** Contains only one outcome. (e.g. rolling a 3 $\rightarrow \{3\}$)
- Compound Event:** Contains more than one outcome. (e.g. rolling an even number $\rightarrow \{2, 4, 6\}$)
- Sure Event:** Occurs with probability 1. (e.g. rolling a number ≤ 6)
- Impossible Event:** Occurs with probability 0. (e.g. rolling a 7 on a fair die)
- Complementary Event:** Event that does not occur.
 If A = getting a 3, then A^c = not getting a 3
 $= \{1, 2, 4, 5, 6\}$

7. Basic Probability Rules

- Probability of any event:
 $0 \leq P(A) \leq 1$
- Probability of a sure event:
 $P(S) = 1$
- Probability of an impossible event:
 $P(\emptyset) = 0$
- Sum of probabilities of all outcomes:
 $\sum P(\text{outcomes}) = 1$

8. Complement Rule

- The probability of an event not occurring.

$$P(A^c) = 1 - P(A)$$

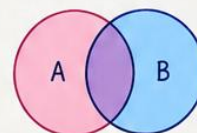
Example:

If $P(A) = 0.3$,
 then $P(A^c) = 1 - 0.3 = 0.7$

9. Addition Rule

- For any two events A and B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



If A and B are mutually exclusive (cannot occur together):

$$P(A \cup B) = P(A) + P(B)$$

10. Multiplication Rule

- For independent events A and B (event B is not affected by A):

$$P(A \cap B) = P(A) \times P(B)$$

- For dependent events:

$$P(A \cap B) = P(A) \times P(B|A)$$

11. Real-Life Examples

- Probability of rain tomorrow = 0.3 (30% chance)
- Probability of a customer buying a product = 0.05
- Probability of defect in a batch = 0.01
- Useful in risk assessment, decision making, A/B testing, fraud detection, etc.

12. Key Takeaways

- ✓ Probability measures how likely an event is.
- ✓ Values range from 0 to 1.
- ✓ Understand sample space and events.
- ✓ Use rules (complement, addition, multiplication) to solve problems.
- ✓ Apply probability to real-world situations.

"Uncertainty is a part of life. Probability helps us navigate it!"

"New information can change everything! That's the power of conditional probability."

Conditional Probability & Bayes' Theorem

Updating beliefs with new information !

"Bayes' Theorem turns data into better decisions." 😊

1. What is Conditional Probability ?

- Conditional probability is the probability of an event A occurring, given that event B has already occurred.
- It tells us how the probability of an event changes when we have additional information.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

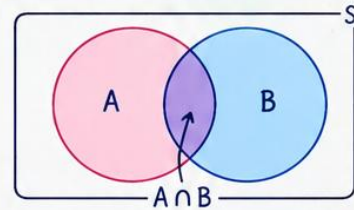
where $P(B) > 0$

2. Intuition

- We restrict our sample space to only those outcomes where B has occurred.
- The probability is recalculated within this smaller sample space.

"It's not the overall probability, but the probability given what we already know."

3. Venn Diagram



$P(A|B) =$
Probability of being in the overlap if we know we are in B.

4. Example 1 : Disease Testing

A test for a disease is 95% accurate (i.e. $P(\text{Test} + | \text{Diseased}) = 0.95$). If a person is known to have the disease, what is the probability that the test is positive?

Solution:

Here, A = Test is positive
B = Person has the disease

So,
 $P(\text{Test} + | \text{Diseased}) = 0.95$

"Knowing the person has the disease increases the probability of a positive test to 95%."

5. Independence

- Two events A and B are independent if the occurrence of one does not affect the probability of the other.
- In that case:

$$P(A|B) = P(A)$$

or equivalently,

$$P(A \cap B) = P(A)P(B)$$

Example:

- A = it rains today
- B = you carry an umbrella (assuming unrelated)
- Then $P(A|B) = P(A)$

6. Bayes' Theorem

- Bayes' Theorem lets us update our belief about an event A based on new evidence B.
- It reverses the conditional probability.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Where:

- $P(A|B)$: Posterior probability (updated belief)
- $P(B|A)$: Likelihood (probability of evidence given A)
- $P(A)$: Prior probability (initial belief)
- $P(B)$: Evidence/Marginal probability

7. Example 2 : Spam Filter

A certain word appears in 70% of spam emails and 10% of normal emails. Assume 20% of all emails are spam. What is the probability that an email is spam given that it contains the word?

Given:

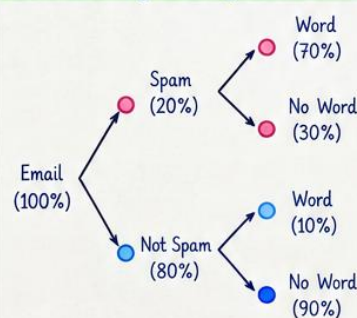
- $P(\text{Word} | \text{Spam}) = 0.70$
- $P(\text{Word} | \text{Not Spam}) = 0.10$
- $P(\text{Spam}) = 0.20 \Rightarrow P(\text{Not Spam}) = 0.80$

Using Bayes' Theorem:

$$\begin{aligned} P(\text{Spam} | \text{Word}) &= \frac{0.70 \times 0.20}{0.70 \times 0.20 + 0.10 \times 0.80} \\ &= \frac{0.14}{0.22} = \mathbf{0.636} \end{aligned}$$

"So, there is a 63.6% chance the email is spam given it contains the word."

8. Tree Diagram (Bayes in action)



"Tree diagrams help visualize conditional probabilities step by step."

9. Key Takeaways

- ✓ Conditional probability looks at the probability of an event given new information.
- ✓ Use $P(A|B) = \frac{P(A \cap B)}{P(B)}$.
- ✓ If events are independent, $P(A|B) = P(A)$.
- ✓ Bayes' Theorem helps update probabilities using new evidence.
- ✓ Useful in real-world problems like medical testing, spam detection, recommendation systems, etc.

"The more we know, the better we estimate!"

Randomness has a pattern... we just need the right tools to see it!

Random Variables & Probability Distributions

From random outcomes to meaningful patterns!

"A random variable turns uncertainty into numbers, and distributions tell their story!"

1. What is a Random Variable?

- A **random variable (RV)** is a numerical value that represents the outcome of a random experiment.
- It assigns a number to each outcome in the sample space.
- It is a function that maps outcomes to real numbers.

$$X : S \rightarrow \mathbb{R}$$

Example:

Let X = number on a fair die roll.
Sample space $S = \{1, 2, 3, 4, 5, 6\}$
 X assigns each outcome its number.

2. Types of Random Variables

Discrete Random Variable

- Takes a countable number of values (e.g. integers).
- Described using a Probability Mass Function (PMF).
- Example:** Number of heads in 3 coin tosses: $X = \{0, 1, 2, 3\}$

Continuous Random Variable

- Takes any value in a continuous range (e.g. real numbers).
- Described using a Probability Density Function (PDF).
- Example:** Height of a person, time to complete a task, temperature, etc.

3. Probability Functions

Probability Mass Function (PMF)

- For discrete random variables.
- $P(X = x)$ gives the probability that X takes value x .
- Must satisfy: $0 \leq P(X = x) \leq 1$
 $\sum P(X = x) = 1$

Example: Fair die
 $P(X = 3) = \frac{1}{6}$

Probability Density Function (PDF)

- For continuous random variables.
- $f(x) \geq 0$ for all x
- Total area under the curve = 1

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

4. Cumulative Distribution Function (CDF)

- Gives the probability that the random variable is less than or equal to a value x .

$$F(x) = P(X \leq x)$$

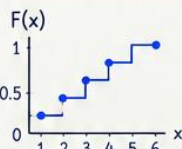
- Properties:
 - $0 \leq F(x) \leq 1$
 - Non-decreasing function
 - $F(-\infty) = 0, F(\infty) = 1$

Example: For a fair die,
 $F(3) = P(X \leq 3) = P(1) + P(2) + P(3)$
 $= \frac{3}{6} = 0.5$

5. Relationship between PMF, PDF and CDF

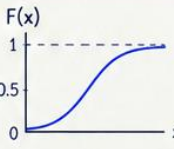
Discrete Case:

$$F(x) = \sum_{t \leq x} P(X = t)$$



Continuous Case:

$$F(x) = \int_{-\infty}^x f(t) dt$$



The PDF is the derivative of the CDF (for continuous random variables):

$$f(x) = \frac{d}{dx} F(x)$$

6. Expected Value (Mean)

- The expected value is the long-run average value of the random variable.

Discrete RV: $E[X] = \sum x P(X = x)$

Continuous RV: $E[X] = \int_{-\infty}^{\infty} x f(x) dx$

Example (Die):

$$E[X] = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = 3.5$$

7. Variance and Standard Deviation

- Variance measures the average squared deviation from the mean.

$$\text{Var}(X) = E[(X - \mu)^2]$$

- Standard deviation is the square root of variance.

$$\sigma = \sqrt{\text{Var}(X)}$$

Example (Die):

$$\text{Var}(X) = \frac{35}{12} \approx 2.92$$

$$\sigma = \sqrt{\frac{35}{12}} \approx 1.71$$

8. Key Points to Remember

- ✓ A random variable assigns numbers to outcomes of a random experiment.
- ✓ Discrete RV $\rightarrow$ PMF, Continuous RV $\rightarrow$ PDF.
- ✓ CDF gives the probability up to a value.
- ✓ Total probability = 1.
- ✓ Expected value is the long-run average.
- ✓ Variance and standard deviation measure spread.
- ✓ Distributions (normal, binomial, etc.) are specific types of random variables.
- ✓ Random variables and their distributions help us model and understand real-world uncertainty.

9. Real-World Examples

- Discrete: Number of customers visiting a store in an hour.
- Discrete: Number of defective items in a batch.
- Continuous: Height, weight, temperature, time, income.
- Continuous: Time to failure of a machine (often follows exponential distribution).

"Different situations, different distributions, but the same idea — modeling uncertainty!"

Different distributions for different stories... Same probability rules!

Important Probability Distributions

Common distributions, real-world use cases and key properties!

"Distributions help us model uncertainty in the real world."

1. Bernoulli Distribution

- Models a single trial with two possible outcomes (success or failure).
- Parameter: p (probability of success)
- $X \sim \text{Bernoulli}(p)$

$$P(X = 1) = p$$

$$P(X = 0) = 1 - p$$

- Mean: $E[X] = p$
- Variance: $\text{Var}(X) = p(1 - p)$

Example: Tossing a coin (head = 1, tail = 0)

2. Binomial Distribution

- Models the number of successes in n independent Bernoulli trials.
- Parameters: n (trials), p (probability of success)
- $X \sim \text{Binomial}(n, p)$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$$

- Mean: $E[X] = np$
- Variance: $\text{Var}(X) = np(1 - p)$

Example: Number of heads in 10 coin tosses.

3. Geometric Distribution

- Models the number of trials until the first success.
- Parameter: p
- $X \sim \text{Geometric}(p)$

$$P(X = k) = (1 - p)^{k-1} p$$

$(k = 1, 2, \dots)$

- Mean: $E[X] = \frac{1}{p}$
- Variance: $\text{Var}(X) = \frac{1 - p}{p^2}$

Example: Number of tosses until the first head.

4. Negative Binomial Distribution

- Models the number of trials to get r successes.
- Parameters: r, p
- $X \sim \text{NegBin}(r, p)$

$$P(X = k) = \binom{k-1}{r-1} p^r (1 - p)^{k-r}$$

$(k = r, r + 1, \dots)$

- Mean: $E[X] = \frac{r}{p}$
- Variance: $\text{Var}(X) = \frac{r(1 - p)}{p^2}$

Example: Number of trials to get 3 heads.

5. Poisson Distribution

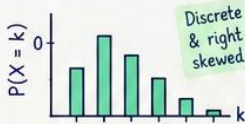
- Models the number of events in a fixed interval of time or space.
- Parameter: λ (average rate)
- $X \sim \text{Poisson}(\lambda)$

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$(k = 0, 1, 2, \dots)$

- Mean: $E[X] = \lambda$
- Variance: $\text{Var}(X) = \lambda$

Example: Number of customer arrivals per hour.



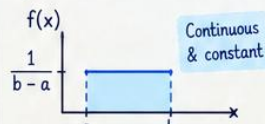
6. Uniform Distribution

- All outcomes are equally likely within a range.
- Parameters: a (min), b (max)
- $X \sim \text{Uniform}(a, b)$

$$f(x) = \frac{1}{b - a} \quad (a \leq x \leq b)$$

- Mean: $E[X] = \frac{a + b}{2}$
- Variance: $\text{Var}(X) = \frac{(b - a)^2}{12}$

Example: Random number generated between 0 and 1.



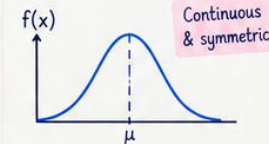
7. Normal Distribution

- Most common continuous distribution (bell-shaped).
- Parameters: μ (mean), σ (std. dev.)
- $X \sim \text{Normal}(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Mean: $E[X] = \mu$
- Variance: $\text{Var}(X) = \sigma^2$

Example: Human height, exam scores, measurement errors.



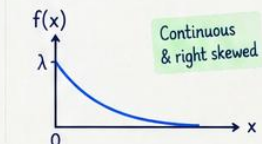
8. Exponential Distribution

- Models the time between events in a Poisson process.
- Parameter: λ (rate)
- $X \sim \text{Exponential}(\lambda)$

$$f(x) = \lambda e^{-\lambda x} \quad (x \geq 0)$$

- Mean: $E[X] = \frac{1}{\lambda}$
- Variance: $\text{Var}(X) = \frac{1}{\lambda^2}$

Example: Time between customer arrivals.



9. Comparison at a Glance

Distribution	Type	Parameters	Mean	Variance	Typical Use Case
Bernoulli	Discrete	p	p	$p(1 - p)$	Single success/failure
Binomial	Discrete	n, p	np	$np(1 - p)$	# of successes in n trials
Geometric	Discrete	p	$1/p$	$(1 - p)/p^2$	Trials until first success
Negative Binomial	Discrete	r, p	r/p	$r(1 - p)/p^2$	Trials until r successes
Poisson	Discrete	λ	λ	λ	# of events in time/space
Uniform	Continuous	a, b	$(a + b)/2$	$(b - a)^2 / 12$	Equally likely outcomes
Normal	Continuous	μ, σ^2	μ	σ^2	Natural phenomena, measurements
Exponential	Continuous	λ	$1/\lambda$	$1/\lambda^2$	Time between events

10. Key Takeaways

- Use discrete distributions for count data and continuous distributions for measurements.
- Choose the distribution based on the real-world context and data characteristics.
- Understand the parameters, mean and variance.
- Many real-world problems (e.g. Poisson, Exponential) are related.
- Normal distribution often arises due to the Central Limit Theorem.
- Selecting the right distribution helps in better modeling, inference and decision-making.

"Same uncertainty... Different distributions... A clearer story!"

"Normal Distribution shows that many real-world phenomena follow a simple pattern!"

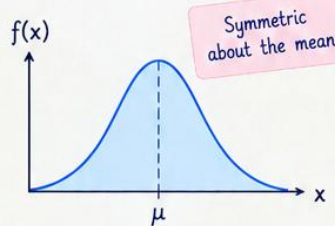
Normal Distribution & Z-Scores

The bell curve that shows up everywhere !

"Z-scores help us see how far a value is from the mean, in standard deviation units !" 😊

1. What is Normal Distribution?

- A normal distribution is a **continuous, symmetric, bell-shaped distribution**.
- Many natural and real-world phenomena (e.g. height, weight, exam scores, measurement errors) follow a normal distribution.
- It is completely determined by two parameters: **mean (μ)** and **standard deviation (σ)**.

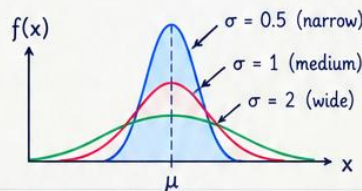


2. Probability Density Function (PDF)

- The probability density function (PDF) of a normal distribution is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- μ : mean (center of the distribution)
- σ : standard deviation (spread)
- The total area under the curve = 1.
- The curve is symmetric about the mean.
- Higher $\sigma \rightarrow$ wider and flatter curve.
- Lower $\sigma \rightarrow$ narrower and taller curve.



3. Z-Score

- A **z-score** tells us how many standard deviations a value x is from the mean.

$$z = \frac{x - \mu}{\sigma}$$

- If $z > 0$, the value is above the mean.
- If $z < 0$, the value is below the mean.
- If $z = 0$, the value is equal to the mean.

Example:

A student scores 85 in a test with mean 70 and standard deviation 10. Find the z-score.

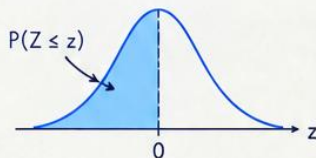
$$z = \frac{85 - 70}{10} = 1.5$$

A z-score of 1.5 means the score is 1.5 standard deviations above the mean.

4. Standard Normal Distribution

- A normal distribution with mean 0 and standard deviation 1.
- Denoted as $Z \sim N(0, 1)$.
- We use the standard normal distribution table (**Z-table**) to find probabilities.

$P(Z \leq z)$ gives the area to the left of z under the standard normal curve.



5. Finding Probabilities (Examples)

Example 1: $P(Z \leq 1.5)$

From Z-table: $P(Z \leq 1.5) \approx 0.9332$

Example 2: $P(Z \geq -1.0)$

Use symmetry: $P(Z \geq -1.0) = P(Z \leq 1.0)$

From Z-table: $P(Z \leq 1.0) \approx 0.8413$

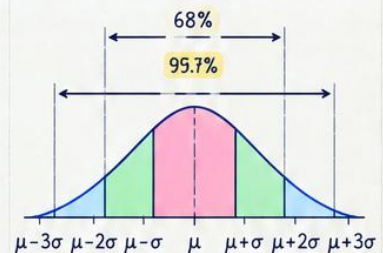
Example 3: $P(-1 < Z < 1)$

$$\begin{aligned} P(-1 < Z < 1) &= P(Z \leq 1) - P(Z \leq -1) \\ &= 0.8413 - 0.1587 \\ &= 0.6826 \end{aligned}$$

About 68.26% of the data lies within 1 standard deviation of the mean.

6. Empirical Rule (68-95-99.7 Rule)

- In a normal distribution:



- ☑ About 68% of the data lies within 1 σ of the mean ($\mu \pm 1\sigma$).

- About 95% lies within 2 σ ($\mu \pm 2\sigma$).
- About 99.7% lies within 3 σ ($\mu \pm 3\sigma$).

7. Common Applications

- Heights and weights in a population
- Test/exam scores
- Measurement errors
- IQ scores
- Financial returns (often assumed normal)
- Quality control in manufacturing
- Many real-world phenomena due to the Central Limit Theorem

"Normal distribution shows up more often than you think!"

8. Key Takeaways

- ✓ Normal distribution is symmetric and bell-shaped.
- ✓ Defined by mean (μ) and standard deviation (σ).
- ✓ Z-score standardizes a value.
- ✓ Use Z-table to find probabilities.
- ✓ Empirical rule gives a quick sense of data spread.
- ✓ Widely used in real-world problems and statistical inference.

"Different data, same bell curve... The world is more normal than we think!" 😊

"A small sample can give us big insights about a population!"

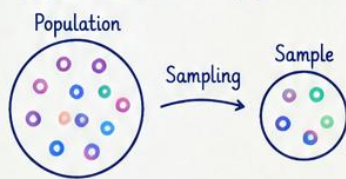
Sampling & Central Limit Theorem

From sample data to population insights !

"The Central Limit Theorem turns randomness into reliability !" 😊

1. What is Sampling ?

- Sampling is the process of selecting a subset of individuals from a larger population to make inferences about the population.
- Helps us save time, cost and resources.
- A good sample should be **representative** of the population.



2. Types of Sampling

Probability Sampling
(every unit has a known chance)

- Simple Random Sampling
- Stratified Sampling
- Cluster Sampling
- Systematic Sampling

Non-Probability Sampling
(chance is unknown)

- Convenience Sampling
- Purposive Sampling
- Quota Sampling
- Snowball Sampling

3. Sampling Distribution

- A sampling distribution is the distribution of a sample statistic (e.g. sample mean) computed from many samples of the same size drawn from a population.
- It shows how much the sample statistic varies from sample to sample.
- Even if the population is **not normal**, the sampling distribution of the mean **tends to be normal** (for large n).

"Different samples, different results... but they follow a pattern !"

4. Sample Statistics vs Population Parameters

Population (Parameter)	Sample (Statistic)
Mean (μ)	Sample mean ($\bar{x}$)
Standard deviation (σ)	Sample standard deviation (s)
Proportion (p)	Sample proportion ($\hat{p}$)
Size (N)	Sample size (n)

5. Standard Error

- Standard error (SE) measures the variability of a sample statistic across different samples.
- For the sample mean:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

- Larger sample size $\rightarrow$ smaller SE.
- Smaller SE $\rightarrow$ more precise estimate.

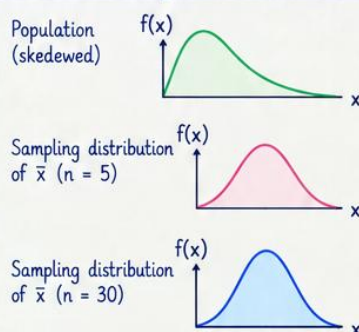
6. Central Limit Theorem (CLT)

- The Central Limit Theorem states that the sampling distribution of the sample mean approaches a normal distribution as the sample size becomes large (usually $n \geq 30$), **regardless of the shape** of the population distribution.

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- Mean of sampling distribution = μ
- Standard deviation = $\frac{\sigma}{\sqrt{n}}$ (standard error)

7. Visualizing the CLT



As sample size increases, the sampling distribution of the mean becomes more normal and less variable !

8. Example

A population has mean $\mu = 50$ and standard deviation $\sigma = 10$. What is the standard error and the distribution of the sample mean for $n = 36$?

Solution:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{36}} = \frac{10}{6} \approx 1.67$$

- Sampling distribution:

$$\bar{x} \sim N(50, 1.67)$$

If we repeatedly take samples of size 36 and calculate the mean, the means will follow a normal distribution with mean 50 and standard deviation 1.67.

9. Key Takeaways

- ✓ Sampling helps us make inferences about a population.
- ✓ A good sample is representative.
- ✓ Sampling distribution shows the variability of a sample statistic.
- ✓ Standard error measures the spread of the sampling distribution.
- ✓ Central Limit Theorem: for large n , $\bar{x}$ is approximately normal, even if the population is not normal.
- ✓ Larger sample size $\rightarrow$ more reliable and precise estimates.

"Big ideas often come from small samples !" 😊

"A confidence interval doesn't give the exact answer, but a range we can trust!"

Confidence Intervals

A range of plausible values for a population parameter

More than a point estimate !

"Same method. Different confidence level. Wider or narrower range !" 😊

1. What is a Confidence Interval ?

- A confidence interval (CI) is a range of values, calculated from sample data, that is likely to contain the true population parameter.
- It is written as:

$$\text{Estimate} \pm \text{Margin of Error}$$

- We are $(1 - \alpha)100\%$ confident that the interval contains the true parameter.

2. Key Components

- Point estimate: Sample statistic ($\bar{x}$ or $\hat{p}$)
- Margin of Error (ME): How much the estimate can vary.
- Confidence Level $(1 - \alpha)$: The percentage of confidence (e.g. 90%, 95%, 99%).
- Critical Value: z^* or t^* value based on the confidence level and sample size.

3. CI for a Population Mean (σ known)

If the population standard deviation σ is known (or n is large), the CI is:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

- $\bar{x}$: sample mean
- σ : population standard deviation
- n : sample size
- z^* : critical value from standard normal distribution

Example: 95% CI for a mean (σ known):
 $\bar{x} = 100$, $\sigma = 15$, $n = 36$, $z^* = 1.96$
 $CI = 100 \pm 1.96 \times \frac{15}{\sqrt{36}} = 100 \pm 4.9$
 $= (95.1, 104.9)$

4. CI for a Population Mean (σ unknown)

If σ is unknown (and n is small), use the t-distribution:

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

- s : sample standard deviation
- t^* : critical value from t-distribution with $(n - 1)$ degrees of freedom

Example: 95% CI for a mean (σ unknown):
 $\bar{x} = 50$, $s = 10$, $n = 16$, $t^* (df = 15) = 2.131$
 $CI = 50 \pm 2.131 \times \frac{10}{\sqrt{16}} = 50 \pm 5.33$
 $= (44.67, 55.33)$

5. CI for a Population Proportion

For a population proportion p , the CI is:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- $\hat{p}$: sample proportion
- n : sample size
- z^* : critical value from standard normal distribution

Example: 95% CI for a proportion:
 $\hat{p} = 0.6$, $n = 100$, $z^* = 1.96$
 $CI = 0.6 \pm 1.96 \sqrt{\frac{0.6 \times 0.4}{100}}$
 $= 0.6 \pm 0.096$
 $= (0.504, 0.696)$

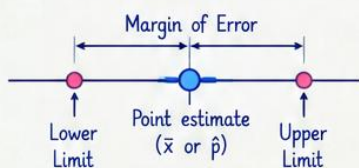
6. Effect of Confidence Level

- Higher confidence level $\rightarrow$ larger critical value $\rightarrow$ wider interval.
- Lower confidence level $\rightarrow$ narrower interval.

Confidence Level	Critical Value (z^*)
90%	1.645
95%	1.960
99%	2.576

More confidence = more uncertainty (range is wider) !

7. Visualizing a Confidence Interval



We are 95% confident that the true population parameter lies somewhere between the lower and upper limits.

8. Assumptions

- Random and representative sample.
- Independent observations.
- For means: population is normal (or n is large, by CLT).
- For proportions: $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$ (normal approximation).

9. Key Takeaways

- ✓ A confidence interval gives a range, not a single value.
- ✓ It depends on the sample statistic, margin of error and confidence level.
- ✓ Higher confidence level $\rightarrow$ wider interval.
- ✓ Use z for large samples/known σ , and t for small samples/unknown σ .
- ✓ Widely used in real-world for estimation and decision making.

"A range today, stronger decisions tomorrow !"

"We don't prove, we find evidence to make a decision!"

Hypothesis Testing Fundamentals

From assumptions to conclusions !

"Same data, different story... It depends on the hypothesis we test!"

1. What is Hypothesis Testing ?

- A statistical method to make a decision about a population parameter based on sample data.
- We test a claim (hypothesis) using evidence from the sample.
- It helps us decide whether the observed result is likely due to random chance or indicates a real effect.

"Use sample data to challenge a claim."

2. Hypotheses

- Null Hypothesis (H_0):** The default claim, usually "no effect" or "no difference".

$$H_0 : \theta = \theta_0$$

- Alternative Hypothesis (H_1):** The claim we want to test (what we suspect).

$$H_1 : \theta \neq \theta_0 \quad (\text{two-tailed})$$

$$H_1 : \theta > \theta_0 \quad (\text{right-tailed})$$

$$H_1 : \theta < \theta_0 \quad (\text{left-tailed})$$

3. Level of Significance (α)

- α is the probability of rejecting the null hypothesis when it is actually true (Type I error).
- Common values: 0.05, 0.01
- A lower α makes it harder to reject H_0 (more evidence needed).

$\alpha = 0.05$ means 5% risk of a false positive.

4. Test Statistic

- A test statistic summarizes the sample data and helps us decide whether to reject H_0 .
- It follows a known distribution (e.g. Z, t, χ^2) under the null hypothesis.

$$Z \equiv \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

- The larger the absolute value of the test statistic, the stronger the evidence against H_0 .

5. p-value

- The p-value is the probability of observing a test statistic as extreme as (or more extreme than) the one calculated, assuming H_0 is true.

- If $p\text{-value} \leq \alpha \rightarrow$ Reject H_0
- If $p\text{-value} > \alpha \rightarrow$ Fail to reject H_0

Small p-value = strong evidence against H_0 .

Example:

If $p\text{-value} = 0.03$ and $\alpha = 0.05$, we reject H_0 (result is statistically significant).

6. Type I and Type II Errors

- Type I Error (α):** Rejecting a true null hypothesis.
- Probability = α (e.g. 0.05)
- Type II Error (β):** Failing to reject a false null hypothesis.
- Probability = β
- Power = $1 - \beta$ (probability of detecting a true effect).

Reality \ Decision	Reject H_0	Fail to Reject H_0
H_0 is true	Type I Error (α)	Correct Decision
H_0 is false	Correct Decision ($1 - \beta$)	Type II Error (β)

7. One-tailed vs Two-tailed Tests

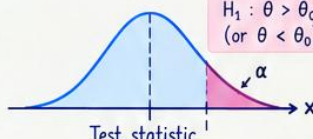
- Two-tailed test:** Tests for any difference ($\neq$).

$$H_1 : \theta \neq \theta_0$$



- One-tailed test:** Tests for a difference in one direction ($>$ or $<$).

$$H_1 : \theta > \theta_0 \quad (\text{or } \theta < \theta_0)$$



8. Steps in Hypothesis Testing

- State the Hypotheses (H_0 and H_1).
- Choose the level of significance (α).
- Select the appropriate test statistic (e.g. Z, t, χ^2).
- Calculate the test statistic and p-value.
- Make a decision (reject or fail to reject H_0).
- Conclude in the context of the problem.

"Statistics gives us evidence, not absolute proof!"

9. Key Takeaways

- ✓ Hypothesis testing helps us make data-driven decisions.
- ✓ H_0 is the default claim; H_1 is what we want to test.
- ✓ Use appropriate test and check assumptions.
- ✓ Compare p-value with α .
- ✓ Understand Type I and Type II errors.
- ✓ Report results with context, not just numbers.
- ✓ A statistically significant result is not always practically significant.

"A significant result is a start for a deeper conversation, not the end!"

"A small p-value doesn't prove the hypothesis, it just gives us evidence against the null!"

p-Values & Statistical Errors

Making the right decision with uncertainty !

"In statistics, we can be wrong... That's why we measure the risk!"



1. What is a p-value ?

- The p-value is the probability of observing a test statistic as extreme as (or more extreme than) the one calculated, assuming the null hypothesis (H_0) is true.
- It measures how compatible the data are with H_0 .
- A small p-value means the observed data are unlikely under H_0 , providing evidence against H_0 .

p-value = $P(\text{data as extreme as observed} \mid H_0 \text{ is true})$

2. How to Interpret p-value ?

- Compare the p-value with the significance level (α) (commonly 0.05).
- If $p\text{-value} \leq \alpha \rightarrow$ Reject H_0 (statistically significant).
- If $p\text{-value} > \alpha \rightarrow$ Fail to reject H_0 (not enough evidence).

Example:

A test gives p-value = 0.03 with $\alpha = 0.05$.

- Since $0.03 < 0.05$, we reject H_0 .
- There is statistically significant evidence against H_0 .

"A small p-value suggests the data is inconsistent with H_0 ."

3. p-value and Evidence Strength

p-value range	Evidence against H_0
$p \geq 0.10$	Little to no evidence
$0.05 \leq p < 0.10$	Weak evidence
$0.01 \leq p < 0.05$	Moderate evidence
$0.001 \leq p < 0.01$	Strong evidence
$p < 0.001$	Very strong evidence

- A smaller p-value provides stronger evidence against H_0 .
- But a large p-value does not prove H_0 is true $\rightarrow$ it only means insufficient evidence to reject H_0 .

4. Type I and Type II Errors

- In hypothesis testing, we can make two types of errors:
- Type I Error (α): Rejecting a true null hypothesis (false positive).
- Type II Error (β): Failing to reject a false null hypothesis (false negative).

Decision	True State of Nature	
	H_0 is True	H_0 is False
Reject H_0	Type I Error (α)	Correct Decision (Power = $1 - \beta$)
Fail to Reject H_0	Correct Decision ($1 - \alpha$)	Type II Error (β)

5. Significance Level (α)

- α is the probability of making a Type I error (rejecting a true H_0).
- Common values: 0.10, 0.05, 0.01.
- A smaller α reduces the chance of Type I error, but may increase the chance of Type II error.

$\alpha = P(\text{Type I Error})$

Example:

If $\alpha = 0.05$, there is a 5% risk of rejecting a true null hypothesis.

6. Power and Type II Error (β)

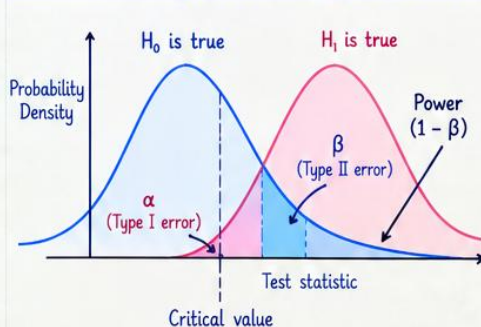
- β is the probability of making a Type II error (failing to reject a false H_0).
- Power = $1 - \beta$ is the probability of correctly rejecting a false H_0 .
- Higher power is better (commonly 0.8 or 80%).

Power = $1 - \beta$

Key point:

- Increase sample size $\rightarrow$ higher power (lower β).
- Larger effect size $\rightarrow$ higher power.
- More variability $\rightarrow$ lower power.

7. Relationship between α , β and Power



8. Key Takeaways

- ✓ p-value tells us how likely our data is under H_0 .
- ✓ Small p-value $\rightarrow$ evidence against H_0 .
- ✓ Large p-value $\rightarrow$ insufficient evidence (don't prove H_0 is true).
- ✓ Type I error (α): false positive.
- ✓ Type II error (β): false negative.
- ✓ Power = $1 - \beta$.
- ✓ Trade-off between α and β .
- ✓ Consider effect size, sample size and context, not just the p-value.

9. Common Misconceptions

- ✗ A small p-value proves the alternative hypothesis.
- ✗ A large p-value proves the null hypothesis is true.
- ✗ p-value tells the probability that the null hypothesis is true.
- ✗ Statistical significance always means practical significance.
- ✗ $\alpha = 0.05$ is a universal rule (for every situation).

"Statistics is not about being certain, it's about managing uncertainty!"



"A statistical test helps us use sample data to make a decision about a population!"

Statistical Tests

Choosing the right test for the right question!

Compare. Decide. Conclude.

"Same logic, different tests... but the goal is always the same - Is the difference real?" 😊

1. Types of Statistical Tests

- Parametric tests**
 - Assume normal distribution (and often equal variances).
 - E.g. z-test, t-test, ANOVA.
- Non-parametric tests**
 - Do not assume normal distribution.
 - E.g. Mann-Whitney U, Wilcoxon, Kruskal-Wallis, Chi-square.
- Used for different purposes:
 - Compare means
 - Compare proportions
 - Test relationships (correlation)
 - Test independence (categorical data)
 - Check goodness of fit

"Choose the test based on your data type, number of groups and assumptions."

2. Z-test (for Population Mean)

- Used when population standard deviation (σ) is known or sample size is large ($n \geq 30$).
- Tests if sample mean differs from a hypothesized population mean (μ_0).

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

- $\bar{x}$: sample mean
- μ_0 : hypothesized mean
- σ : population standard deviation
- n : sample size

Example:

A company claims the average delivery time is 5 days ($\sigma = 1.5$). A sample of 36 orders has a mean of 5.5 days. Test at $\alpha = 0.05$.

$$Z = \frac{5.5 - 5}{1.5/\sqrt{36}} = 2.0$$

Since $p < 0.05$, we reject H_0 .

3. t-test (for Population Mean)

- Used when population standard deviation is unknown (most common).
- Assumes data is approximately normally distributed.
- Used for small samples ($n < 30$) or when σ is unknown.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

- $\bar{x}$: sample mean
- μ_0 : hypothesized mean
- s : sample standard deviation
- n : sample size
- Degrees of freedom (df) = $n - 1$

Example:

A sample of 15 employees has a mean salary of 52K ($s = 8K$). Test if the mean is different from 50K.

$$t = \frac{52 - 50}{8/\sqrt{15}} = 0.97$$

If $p > 0.05$, fail to reject H_0 .

4. Tests for Two Means

Test	When to use	Statistic
Independent t-test	Compare means of two independent groups	t
Paired t-test	Compare means of the same group (before vs after)	t
Welch's t-test	When variances are unequal	t

"Use paired t-test for before-after or matched data."

5. ANOVA (Analysis of Variance)

- Used to compare means of 3 or more groups.
- Tests the null hypothesis that all group means are equal.
- Assumes normality and equal variances across groups.

$$F = \frac{\text{Between-group variability}}{\text{Within-group variability}}$$

Hypotheses:

- $H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$
- H_1 : At least one mean is different.

Note:

- If ANOVA is significant, use post-hoc tests (e.g. Tukey) to find which groups differ.
- For non-normal data, use Kruskal-Wallis test.

6. Tests for Proportions

Test	When to use	Statistic
One-sample proportion test	Test if a single proportion equals a known value	Z
Two-proportions z-test	Compare proportions of two independent groups	Z

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$$

- $\hat{p}$: sample proportion
- p_0 : hypothesized proportion
- n : sample size

7. Non-parametric Tests

Test	When to use	Alternative to
Mann-Whitney U	Compare 2 independent groups	Independent t-test
Wilcoxon Signed-Rank	Compare 2 related groups	Paired t-test
Kruskal-Wallis	Compare 3+ groups	ANOVA
Chi-square test	Test independence of categorical variables	-

Use non-parametric tests when data is not normally distributed or sample size is small.

8. Chi-square Test

- Used for categorical data.
- Common uses:
 - Test independence (association between variables)
 - Goodness of fit (compare observed vs expected frequencies)

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- O : observed frequency
- E : expected frequency
- Degrees of freedom = $(r - 1)(c - 1)$

9. Key Takeaways

- ✓ Choose the test based on data type, number of groups and assumptions.
- ✓ Check assumptions (normality, equal variance, independence).
- ✓ Use parametric tests when assumptions are met, otherwise use non-parametric.
- ✓ $p\text{-value} < \alpha \rightarrow$ reject H_0 (statistically significant).
- ✓ $p\text{-value} \geq \alpha \rightarrow$ fail to reject H_0 .
- ✓ Statistical significance $\neq$ practical significance.
- ✓ Always consider context and effect size.

"A test gives a p-value, but domain knowledge gives the real meaning!" 😊

"Not all relationships are causal, but they often tell a story!"

Correlation & Covariance

Understanding the relationship between variables !

"Correlation measures strength, Covariance shows direction !" 😊

1. What is Covariance ?

- Covariance measures the direction of the linear relationship between two variables.
- If both variables tend to increase (or decrease) together → covariance is **positive**.
- If one variable increases while the other decreases → covariance is **negative**.
- If there is no linear relationship → covariance is zero (or close to zero).

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- x_i, y_i : observations
- $\bar{x}, \bar{y}$: sample means
- n : sample size

2. What is Correlation ?

- Correlation measures the strength and direction of the linear relationship between two variables.
- It is a standardized version of covariance, so it is unitless and always between -1 and +1.

$$r = \frac{\text{Cov}(X, Y)}{s_X / s_Y}$$

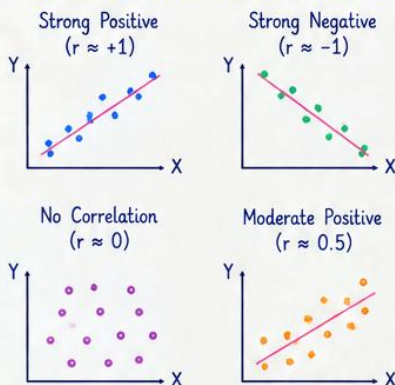
- s_X, s_Y : sample standard deviations of X and Y
- r : Pearson correlation coefficient

Correlation tells us how strongly variables move together, but not that one causes the other.

3. Interpretation

Value of r	Interpretation
+1	Perfect positive linear relationship
0.7 to 0.9	Strong positive correlation
0.3 to 0.7	Moderate positive correlation
0 to 0.3	Weak positive correlation
0	No linear correlation
-0.3 to 0	Weak negative correlation
-0.7 to -0.3	Moderate negative correlation
-0.9 to -0.7	Strong negative correlation
-1	Perfect negative linear relationship

4. Visualizing Correlation



5. Example Calculation

Consider the following data for hours studied (X) and test score (Y):

Student	X (Hours)	Y (Score)
1	2	50
2	4	65
3	6	70
4	8	85
5	10	95

Using the formulas:

- $\text{Cov}(X, Y) = 187.5$
- $r = 0.98$

Interpretation: Strong positive correlation between study hours and test scores.

6. Important Points

- Covariance depends on the **units** of the variables, correlation is **unitless**.
- Correlation is always between **-1 and +1**.
- Correlation captures only linear relationships.
- A correlation of 0 does **not necessarily** mean no relationship (could be non-linear).
- Correlation is sensitive to **outliers**.
- High correlation does not imply **causation**.
- Use scatter plots to visualize the relationship.

7. Covariance vs Correlation

Aspect	Covariance	Correlation
Measures	Direction of linear relationship	Strength and direction of linear relationship
Range	$(-\infty, +\infty)$	$[-1, +1]$
Units	Depends on units of variables	Unitless
Interpretation	Positive / Negative / Zero	Strength (weak/strong) and direction
Use case	Useful in modeling (e.g. portfolio theory)	Easier to interpret and compare

8. Key Takeaways

- ✓ Create scatter plots before calculating correlation.
- ✓ Use correlation to understand the strength of linear relationship.
- ✓ Remember: correlation $\neq$ causation.
- ✓ Be cautious of outliers and non-linear relationships.
- ✓ Choose the right measure based on your analysis goal.

"Correlation finds patterns, Covariance shows the direction, But critical thinking finds the truth !" 😊

"Data speaks louder when you find the right model!"

Regression & Statistical Modeling

Model the relationship. Make better predictions!

"A model is a simplified version of reality, not reality itself!"

1. What is Regression?

- Regression is a statistical method used to model the relationship between a dependent variable (Y) and one or more independent variables (X).
- It helps us understand the relationship and make predictions.
- Most common type: Linear Regression.

"Find the line that best explains the data."

2. Simple Linear Regression

- Models the relationship between one independent variable (X) and a dependent variable (Y).
- Equation:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- β_0 : intercept (value of Y when X = 0)
- β_1 : slope (change in Y for a unit change in X).
- ϵ : error term (captures variation not explained by the model)

3. Multiple Linear Regression

- Models the relationship between multiple independent variables and a dependent variable.
- Equation:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$$

- $X_1, X_2, \dots, X_p$: independent variables
- $\beta_0, \beta_1, \dots, \beta_p$: model coefficients
- ϵ : error term

Example:

Predict house price using area, number of rooms and location.

$$\text{Price} = \beta_0 + \beta_1 (\text{Area}) + \beta_2 (\text{Rooms}) + \beta_3 (\text{Location}) + \epsilon$$

4. Fitting the Model (Least Squares)

- We estimate the coefficients by minimizing the sum of squared errors between actual and predicted values.

$$\text{Minimize: } \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



The best fit line minimizes the total error!

5. Model Evaluation Metrics

Metric	Meaning
R^2	Proportion of variance in Y explained by the model (0 to 1).
Adjusted R^2	R^2 adjusted for number of predictors.
MSE	Mean Squared Error (average of squared errors).
RMSE	Square root of MSE (in same unit as Y).
MAE	Mean Absolute Error (average of absolute errors).

Higher R^2 is better.
Lower MSE, RMSE and MAE are better.

6. Assumptions (for Linear Regression)

1. Linearity: Relationship between X and Y is linear.
2. Independence: Observations are independent.
3. Homoscedasticity: Constant variance of errors (equal spread).
4. Normality: Errors are normally distributed.
5. No multicollinearity: Independent variables are not highly correlated.

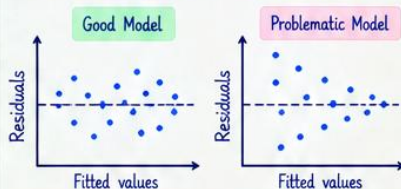
Check assumptions using residual plots, histograms and VIF (for multicollinearity).

7. Residual Analysis

- Residual = Actual value - Predicted value

$$e_i = y_i - \hat{y}_i$$

- Residual plots help us check assumptions (like linearity and constant variance).



Random pattern → assumptions likely satisfied.

Pattern (e.g. funnel) → violation of assumptions.

8. Types of Regression Models

Type	Use Case
Linear Regression	Continuous outcome, linear relationship
Polynomial Regression	Non-linear relationship (uses polynomial terms)
Ridge Regression	Handles multicollinearity (L2 regularization)
Lasso Regression	Feature selection (L1 regularization)
Logistic Regression	Binary outcome (classification)
Poisson Regression	Count data (e.g. number of events)

9. Key Takeaways

- ✓ Regression helps us understand relationships and make predictions.
- ✓ Interpret coefficients in context (units matter!).
- ✓ Always check model assumptions.
- ✓ Use appropriate evaluation metrics.
- ✓ Consider regularization (Ridge/Lasso) when needed.
- ✓ Residual analysis is essential.
- ✓ Choose the right model for the problem.

"All models are wrong, but some are useful."
- George E. P. Box

"Good Data + Right Statistics = Better Decisions!"

Statistics for Data Science

Master Cheat Sheet

"Same concepts. Bigger insights. From data to decisions!"

1. Descriptive Statistics

- Summarize and understand data.
- Measures of central tendency: Mean ($\bar{x}$), Median, Mode
- Measures of variability: Range, Variance (s^2), Standard Deviation (s)
- Position: Quartiles (Q_1, Q_2, Q_3), Percentiles, IQR
- Detect outliers: IQR rule $\rightarrow [Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}]$

Descriptive statistics gives a snapshot of your data.



2. Probability Fundamentals

- $0 \leq P(A) \leq 1$
- Complement: $P(A^c) = 1 - P(A)$
- Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Multiplication rule: $P(A \cap B) = P(A)P(B|A)$
- Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- Bayes' theorem: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

Probability quantifies uncertainty in the real world.

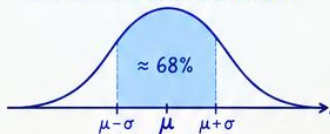


3. Important Distributions

Distribution	Use Case	Key Formula / Property
Bernoulli	Single trial (success/failure)	$P(X = 1) = p$
Binomial	# of successes in n trials	$X \sim \text{Bin}(n, p)$ $E[X] = np$
Poisson	Count of events in time/space	$X \sim \text{Pois}(\lambda)$ $E[X] = \text{Var}[X] = \lambda$
Uniform	All outcomes equally likely	$X \sim U(a, b)$ $E[X] = (a + b) / 2$
Normal	Many real-world data (continuous)	$X \sim N(\mu, \sigma^2)$
Exponential	Time between events	$X \sim \text{Exp}(\lambda)$ $E[X] = 1/\lambda$

4. Normal Distribution & Z-Scores

- Bell-shaped, symmetric distribution.
- Standard normal: $Z \sim N(0, 1)$
- Z-score: $Z = \frac{X - \mu}{\sigma}$
- Use Z to find probabilities and compare values from different distributions.



Z-scores tell us how many standard deviations a value is from the mean.

5. Sampling & Central Limit Theorem

- A sample is a subset of the population.
- Sampling methods: Simple random, stratified, cluster, systematic.
- Central Limit Theorem (CLT): For large n , the sampling distribution of the sample mean ($\bar{x}$) is approximately normal, regardless of the population distribution.

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- Mean of sampling distribution = μ
- Standard error = $\frac{\sigma}{\sqrt{n}}$



Larger sample size $\rightarrow$ more reliable and stable estimates.

6. Confidence Intervals

- A confidence interval (CI) gives a range of plausible values for a population parameter.
- General form: Estimate $\pm$ (Critical value $\times$ SE)
- For mean (σ known): $\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
- For mean (σ unknown): $\bar{x} \pm t_{\alpha/2, df} \frac{s}{\sqrt{n}}$
- Common confidence levels: 90% (1.645), 95% (1.96), 99% (2.576)

A 95% CI means: if we repeated the sampling many times, 95% of intervals would contain the true parameter.

7. Hypothesis Testing

- Test a claim about a population parameter.
- Steps:
 - State H_0 and H_1
 - Choose significance level (α)
 - Select appropriate test statistic
 - Calculate test statistic and p-value
 - Make a decision (reject or fail to reject H_0)

We don't prove, we find evidence to make a decision.



8. p-Values & Statistical Errors

- p-value: Probability of observing a test statistic as extreme as (or more than) the one calculated, assuming H_0 is true.
- If p-value $\leq \alpha \rightarrow$ Reject H_0
- If p-value $> \alpha \rightarrow$ Fail to reject H_0
- Small p-value $\Rightarrow$ strong evidence against H_0 .

Reality	Decision	Result	Term
H_0 true	Reject H_0	Type I Error (False Positive)	α
	Fail to reject H_0	Correct Decision	$1 - \alpha$
H_0 false	Reject H_0	Correct Decision (Power)	$1 - \beta$
	Fail to reject H_0	Type II Error (False Negative)	β

9. Statistical Tests (Quick Guide)

Test	Use Case	Assumptions
Z-test	Mean (σ known)	Normal or large n
t-test	Mean (σ unknown)	Normal or large n
Paired t-test	Before-after / matched data	Differences $\sim$ normal
ANOVA	Compare 3+ means	Normal, equal variances
Chi-square	Categorical data (goodness of fit / independence)	Independent observations
Correlation	Linear relationship	Linear, no strong outliers
Regression	Predict Y using X	Linearity, independent errors, constant variance

10. Correlation & Covariance

- Covariance: Direction of linear relationship.
$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$
- Correlation: Standardized covariance.
$$r = \frac{\text{Cov}(X, Y)}{s_x s_y} \quad (-1 \leq r \leq 1)$$
- $r > 0$: positive, $r < 0$: negative, $r = 0$: no linear relationship

11. Regression & Modeling

- Simple Linear Regression:
$$Y = \beta_0 + \beta_1 X + \epsilon$$
- Multiple Linear Regression:
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$$
- Assess model: R^2 , Adjusted R^2 , RMSE, MAE
- Check assumptions: linearity, independence, homoscedasticity, normality of residuals.
- Use regularization (Ridge/Lasso) when needed.

12. Key Takeaways

- Understand the problem and choose the right method.
- Check assumptions before applying any test/model.
- Use visualizations to explore and validate.
- Interpret results in context (business + data).
- Statistics is about uncertainty, not absolute truth.
- A significant result is a starting point for a deeper conversation.
- Combine domain knowledge with statistical insights.

"Better questions. Brighter insights. Bigger impact."